



Google Classroom

Google has started previewing its latest attempt to empower teachers and students with a new teaching aid <http://dgit.in/1kcwUsv>



Oculus Rift inspires Samsung

Samsung is working on its own version of the Oculus Rift, and is said to have already shipped prototypes of "Morpheus" <http://dgit.in/morpheusOR>

Projectors

HOW WE TESTED

Since we deal with variable light sources on the different projectors, a bit of prep work is needed to ensure uniform test conditions. We tested the projectors in a dark room with no reflective surfaces (glass, mirrors, etc). A high gain white screen measuring 6-feet-by-4-feet (with a black border) was suspended 2 metres from where the projectors were kept on a table at a height of 3 feet from the floor. We used a Minolta light meter to measure the light coming from the projectors -- all testing was done after ensuring there was next to no ambient light in the room (before firing up every projector, we ensured the meter showed E or Error, which means not enough light to register).

Performance

We hooked up all the projectors to a standard rig with an HDMI port. The test projectors were connected to the rig via an HDMI cable to run video files and benchmarks off it. To ensure uniformity, all projectors were restored to factory settings before benchmarking

their individual performance. Prime among the synthetic tests was DisplayMate's suite of industry standard display benchmark (for more info visit www.displaymate.com). The program lets you test a projector's image for various things like distortion, sharpness, brightness, colour, pixel density, etc. It's a comprehensive test, used to test not only projectors but TVs and monitors, and we've included only a small snapshot of it in the comparison table. Brightness was calculated by displaying a full white image and taking a reading at the screen's surface, with the light meter's sensitive bulb facing the full glow of the projector's uninterrupted light. We also ran PixPerAn's readability test to measure ghosting and text clarity.

Apart from these objective tests, we watched movies like Avatar, Batman Begins, Up, Transformers, Sin City, and Quantum Of Solace -- all HD 1080p MKV files. We looked for ghosting, inconsistencies in colour, colour flashes and rainbow effects which usually occur while watching fast-paced, colourful

cinematic visuals. For the projectors sporting onboard speakers, we tested their sound quality, too -- but, honestly, they were quite bad, and if you're planning to spend close to a lakh on a projector without investing in a good system, you have another thing coming. We also checked each projector's startup time and its peak operating temperature (near its exhaust vent) through a digital thermometer.

Features and Design

Parameters to differentiate between the projectors -- like type and number of connectivity ports, native resolution, warranty, remote control, 3D support, etc., and specifications that didn't have a direct impact on their performance -- were tabulated under the Features sheet. The projector's build quality, ease of setup, quality of on-screen interface, and other similar salient attributes were tabulated under the Design sheet. After applying our formula to determine winners, we came up with the Best Performer, Best Buy and Editor's Pick projectors. We're sure you'll like them.

qualities of performance that seem to be a hallmark of the manufacturer. This projector had exaggerated blues to the point that blue dominated the colour in almost every frame, wherever it was present. In movies where blues were relatively less -- Sin City, for example -- we found the projector to be pleasantly entertaining and displaying crisp, vivid visuals. But Avatar, let's just say watching it was a bit of an eye sore on this projector. All this is of course in the Standard mode -- Cinema or movie mode, balances out the colours a bit but not without affecting image quality, making the final projected output a bit washed out and relatively dull.

On-board audio of the projector is in line with other BenQ projectors we tested,

with nothing special to write home about. The speakers are loud, enough to fill a medium sized room with audio, but it isn't clear or noise-free beyond the 60-65 per cent volume levels. As we said earlier, no self-respecting projector owner should settle for on-board audio bundled in these projectors, because they're just shoddy and be used as a last resort at all times.

The remote is larger than we expected and has a spring-loaded kickstand under its front lip. We liked the fact that the upper surface of the projector had labels indicating the position of the various connectivity ports, making cable management a breeze since we didn't have to turn the projector over every time we wanted to plug in something else. A nice touch.

BenQ MH680

This is a nice, small and petite projector, smaller than the BenQ W1400 and W1070 in fact, and it comes with its own carry bag. It looks different from the other BenQ projectors as well, owing to that excessive dash of dark blue. The look and feel of the projector is nice, with a pleasant two-toned finish. The top of the lens has a nice, textured, faux-wood panel finish which is a nice touch compared to other lackadaisical projectors.

It's extremely easy to setup. On a completely white image,



BenQ MH680

a moiré pattern with very fine lines was visible across 2/3rd of the projected image and largely to the right side of the projection. Also its top edge had a greenish tinge to it (barely perceptible) and the entire image wasn't neutral, but appeared warm with a slight pink area towards the top right of the frame.

The projector displays extremely sharp and vivid images. The test images in DisplayMate were extremely bright and well portrayed. However, its oversaturation effects were visible when we started watching movies. Avatar highlighted the exaggerated blues in the movie. However, we'd still rate the image and video quality of the MH680 at par of the W1070 (or even slightly better), and both are definitely better than